

Very important

Topic: Pointer to array (Array pointer)

18 Jan 2022

★ Revision

int x=2, *p1=&x, **p2=&p1;

→ printf("%p %p %p", &x, p1, p2);
// 400 400 800

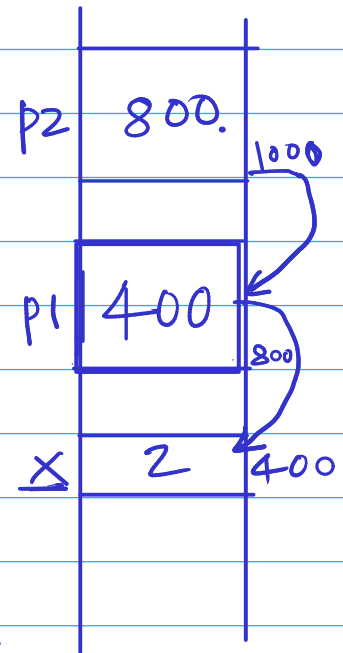
printf("%p %p %p", &x, &p1, &p2);
// 400 800 1000

printf("%d %d %d", x, *p1, **p2);
// 2 2 2

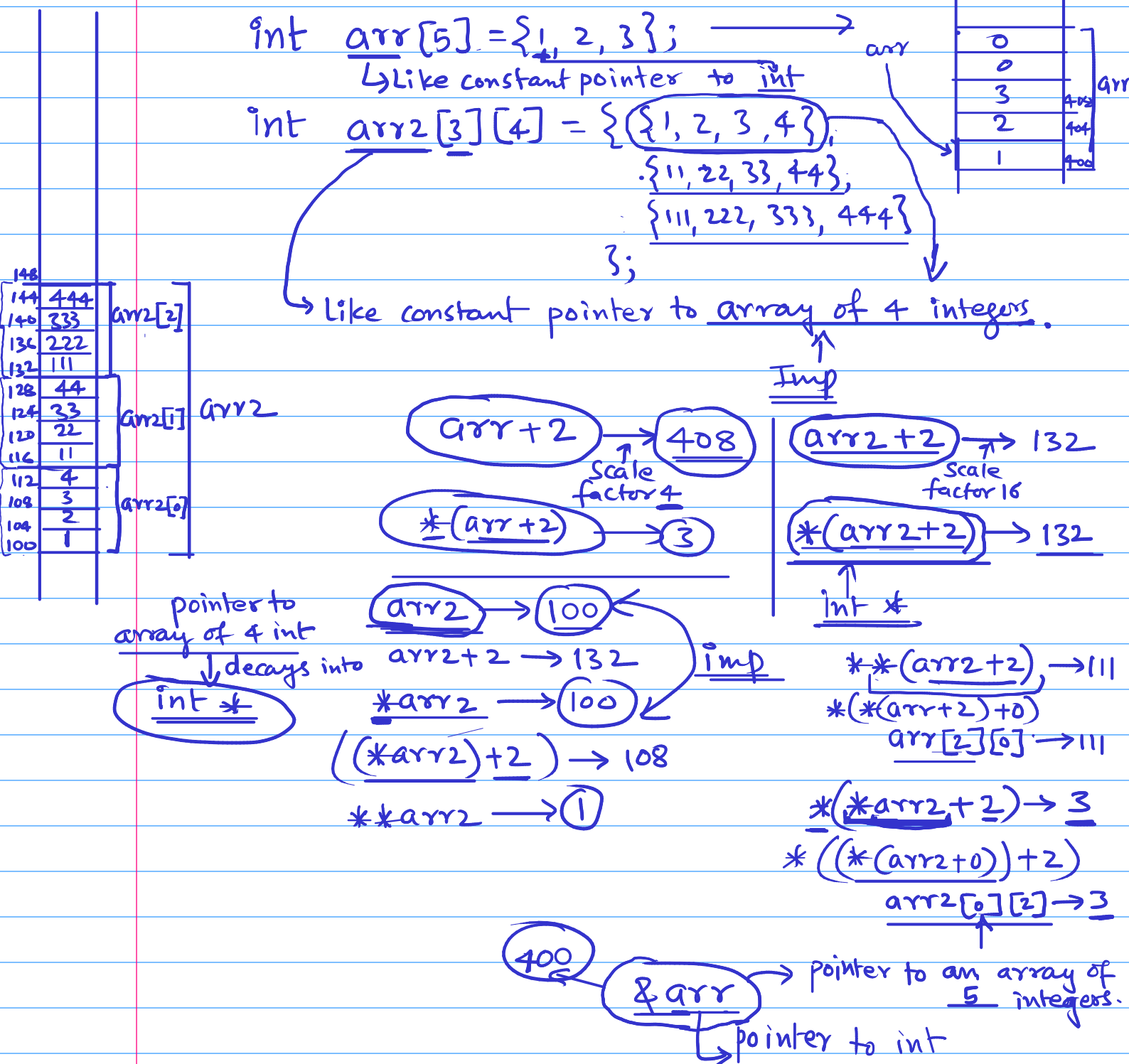
printf("%p %p %p", &x, p1, *p2);
// 400 400 400 ← Imp

p2 — pointer to int pointer
(*p2) and p1 — pointer to int
**p2 and *p1 and x — int

X int ***p3 = &p1; // Invalid
✓ int ***p3 = &p2; // valid



★ Pointer to an array (Array pointer)



```
int arr[5] = {1, 2, 3};
printf("%p", arr);
//400
printf("%p", &arr);
//400
printf("%.d", *arr);
//1
```

```
printf("y.p", arr+1);
//404
printf("%p", (&arr)+1);
//420
printf("%.p", *(&arr+1));
//420
```

0	
0	
3	
2	
1	400

int (*ptr)[5]; → pointer to an array of 5 int.

How much memory will get allocated for ptr?
same as pointer.
8 bytes

✓ ptr = &arr;

int (*ptr2)[4];

X ptr2 = &arr;

✓ ptr2 = arr2;